12-14 第2篇

阅读理解（说明文） Why Do Geese Fly in a "V" Shape?

① Every year in late autumn, somewhere over the fields of northern China, a small group of children will look up at the clear cold sky and see a moving **dark V** high above them — a flock of wild geese flying south for the winter. Twenty or thirty birds, perfectly spaced, one bird at the front of the V, two lines spreading backward behind her like a pair of open scissors. The sight is so common that most of us never really ask the question: **Why a V? Why not a straight line, or a neat square, or a simple cloud of birds?** The answer, it turns out, is one of the quiet small wonders of bird biology — and it is mostly about **saving energy**.

② **Finding one: flying is very, very expensive.** A Canada goose, weighing about **4 kilograms**, burns roughly **8 times more energy per minute when flying than when standing still on a lake**. On a long autumn migration — the bar-headed goose, for example, flies **5,000 kilometres** from Mongolia to India each year — every gram of energy saved matters. If the bird cannot save energy on the way, she will not arrive alive. Evolution, over millions of years, has therefore made geese very good at one thing: **finding ways to fly for less**.

③ **Finding two: every flapping wing leaves behind a small invisible stair.** When a bird's wing pushes the air down, the air does not simply disappear: it curls into two small rotating rivers of air called **wingtip vortices**, one off the end of each wing. Directly behind the wingtip, the air actually moves **upward** — a small invisible rising "stair" in the sky. If a second bird flies in exactly the right spot behind her — not directly behind, but slightly to the side — her own wings can catch this upward-moving air, and she gets a **free push upward**. Her own muscles can, for that moment, do less work. The clever V-shape is simply the **geometry that places every bird except the very first one in the upward air of the bird in front of her**.

④ **How much energy does this actually save?** In **2001**, a French research team (Weimerskirch and colleagues, at the CNRS lab in Chizé) fitted tiny heart-rate monitors to eight trained **white pelicans** flying in V-formation behind a small motorboat. The pelicans flying **behind** another bird had heart rates **11% to 14% lower**, and flapped their wings **about 7% less often**, than a pelican flying alone. A later 2014 study at the University of London, using tiny GPS-loggers on **14 northern bald ibises** migrating in V-formation, went further: the birds were so careful about timing that each wingbeat was **synchronised within 0.15 seconds** with the bird in front, placing each wing exactly inside the upward air stair. These are not accidents. These are **small living aerodynamic calculations**, made by a brain the size of a walnut, twenty times per second.

⑤ **Finding three: the leader at the front of the V has the hardest job.** The front bird gets no upward help; she flies in pure, untouched air. Her heart rate is higher. Her wings beat harder. So geese, quietly and often, **rotate the leader**. The same 2014 ibis study, using GPS trails, found that a single ibis flew at the front of the V for **only about 32% of its time in the air** before another bird moved up and took over. There is no shouted order, no cruel test. The flock, somehow, shares the hardest seat in the sky.

⑥ **Finding four: the V is also a pair of kind eyes.** Research at the University of California, Davis (2018) suggested a second quiet reason for the V: **every bird in the formation can see every other bird** out of the corner of its eye. In a straight line, only two birds can see each other clearly. In a tight square, inside birds cannot see outer birds. In a V, the angle is about **45°**, which keeps each bird visible to all the others. This matters for small steering signals — a tiny head-turn, a small dip of the wings — that warn the flock about wind, cold air, a rising mountain, or a hawk in the distance. **The V is, quietly, a form of silent group conversation.**

⑦ **Do other birds use the V?** Yes — but mostly the heavy, long-wing, long-distance migrators. **Geese, swans, pelicans, cranes, ibises** all use V-formations in some form. **Small songbirds** (sparrows, swallows) do not: their wings are too small and their vortices too weak for the saving to matter. Tiny birds usually fly in **loose, fast-changing clouds** called **murmurations**, which serve different purposes (protection from hawks, not energy saving). So the V is a **heavy bird's** invention.

⑧ **What about the young geese?** A Chinese study from the Qinghai Lake Nature Reserve (2023, 132 young bar-headed geese fitted with light trackers) found that **first-year geese make about 3 to 5 extra metres of position error** compared to adult geese — they do not catch the upward-stair as perfectly. The young goose's heart rate is correspondingly higher on her first migration. But by the following spring, after **one full round trip of 10,000 km**, the same young goose's position error is **down to adult levels**. Flying in V, scientists quietly argue, is therefore **not fully instinct** — it is **instinct plus a careful year of learning**, taught by the older birds.

⑨ **The quiet lesson.** The next time you see a long dark V moving slowly across a cold autumn sky, remember: you are not watching a neat pattern. You are watching **one of the cleverest small engineering solutions in nature** — a solution made without words, shared between twenty careful birds, updated twenty times per second, that saves perhaps **13% of their energy** on a 5,000-km journey. You are watching a small, ancient, almost-invisible piece of cooperation: every bird helping, for a few minutes, every other bird. And when the leader at the front of the V tires, somebody quietly moves up, takes her place, and lets her rest.

注释： migration 迁徙；vortex/vortices 涡流；aerodynamic 空气动力学的；synchronise 同步；murmuration 鸟群同步群飞

根据短文内容，选择最佳答案。

1. According to Paragraph ②, why is energy-saving so important for geese?

A. Because geese love lazy lives and do not like flying.
B. Because flying is very expensive — a Canada goose weighing about 4 kilograms burns roughly 8 times more energy per minute when flying than when standing still on a lake, and on a long autumn migration (e.g. the bar-headed goose flies 5,000 km from Mongolia to India each year), every gram of energy saved matters; if the bird cannot save energy on the way, she will not arrive alive, so evolution has made geese very good at finding ways to fly for less.
C. Because geese cannot eat during migration, so they do not need any energy.
D. Because geese only fly at night and the air is warmer.

2. According to Paragraph ③, what exactly causes the V-shape to save energy?

A. The V-shape looks pretty from the ground, and ground-based viewers give the geese food.
B. The V-shape protects birds from the sun.
C. The V-shape helps geese make a loud noise that scares hawks.
D. When a bird's wings push air down, the air curls into two wingtip vortices, and directly behind the wingtip the air actually moves upward — a small invisible rising "stair" in the sky; if a second bird flies slightly to the side and in exactly the right spot behind her, its wings catch this upward-moving air and get a free push upward, so the V-shape is simply the geometry that places every bird except the very first one inside the upward air of the bird in front.

3. What did the 2001 French pelican study and the 2014 London ibis study each measure and find?

A. The French study found the pelicans had heart rates 50% lower flying behind another bird; the London study found ibises did not care about timing.
B. The French study found zero energy savings; the London study found ibises only flew at night.
C. The French team (Weimerskirch, CNRS Chizé) fitted tiny heart-rate monitors to 8 trained white pelicans flying in V behind a motorboat and found those flying behind another bird had heart rates 11%—14% lower and flapped about 7% less often than a solo pelican; the 2014 University of London study used tiny GPS-loggers on 14 northern bald ibises and found each wingbeat was synchronised within 0.15 seconds with the bird in front — the wing being placed exactly inside the upward air "stair".
D. Both studies were about songbirds, not geese.

4. According to Paragraphs ⑤ and ⑥, what are two quiet non-energy reasons the V-shape matters, and what do they show about the flock?

A. The leader is forced to stay in front forever because she is the strongest; the V also helps birds hide from view.
B. The front bird gets no upward help and has a higher heart rate and harder wing-beats, so geese quietly rotate the leader (the 2014 ibis study using GPS found a single ibis flew at the front only about 32% of its air time before another bird took over — no shouted order, no cruel test); the V also keeps every bird visible to every other bird at about 45° (UC Davis 2018), which matters for small steering signals (head-turns, wing-dips) warning the flock about wind, cold air, a rising mountain or a hawk — so the V is a silent form of group conversation, and the flock quietly shares the hardest seat in the sky.
C. The leader eats first and other birds get no food; the V is decorative only.
D. The V only exists so that hawks cannot see the flock at all.

5. What is the main idea of the final paragraph?

A. Children should catch geese to learn about flying.
B. The V-shape is a meaningless accident and no one understands it.
C. All birds fly in V-shape, including sparrows and swallows.
D. The V-shape high in a cold autumn sky is not a neat pattern but one of nature's cleverest small engineering solutions — a wordless, shared, constantly-updated system that saves perhaps 13% of the geese's energy on a 5,000-km journey, and is a small ancient piece of cooperation in which every bird helps every other bird, and when the leader tires, somebody quietly moves up and takes her place so she can rest.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段飞行代价
【解析】"8 倍能耗 / 5000 公里 / 节省每克能量才能到达"直引。A、C、D 均错。故选 B。

2.
【答案】D
【题型】细节理解题
【定位】第③段翼尖涡
【解析】"下压 / 涡流 / 翼尖后方上升气流 / V 字几何"直引。A、B、C 均错。故选 D。

3.
【答案】C
【题型】细节理解题
【定位】第④段两项研究
【解析】"11-14% 心率下降 / 7% 挥翅减少 / 0.15 秒同步"直引。A、B、D 均错。故选 C。

4.
【答案】B
【题型】推理判断题 / 细节理解题
【定位】第⑤⑥段
【解析】"32% 前位 / 45° 互相可见 / 无声沟通"推理。A、C、D 均偏离。故选 B。

5.
【答案】D
【题型】主旨大意题
【定位】第⑨段结尾
【解析】"13% 能量 / 无声合作 / 领头疲劳有鸟替换"主旨。A、B、C 均偏离。故选 D。